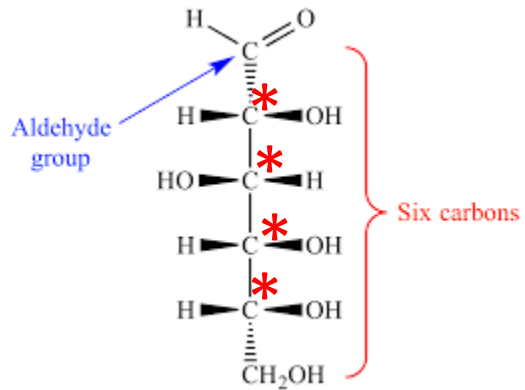


Carbohydrates

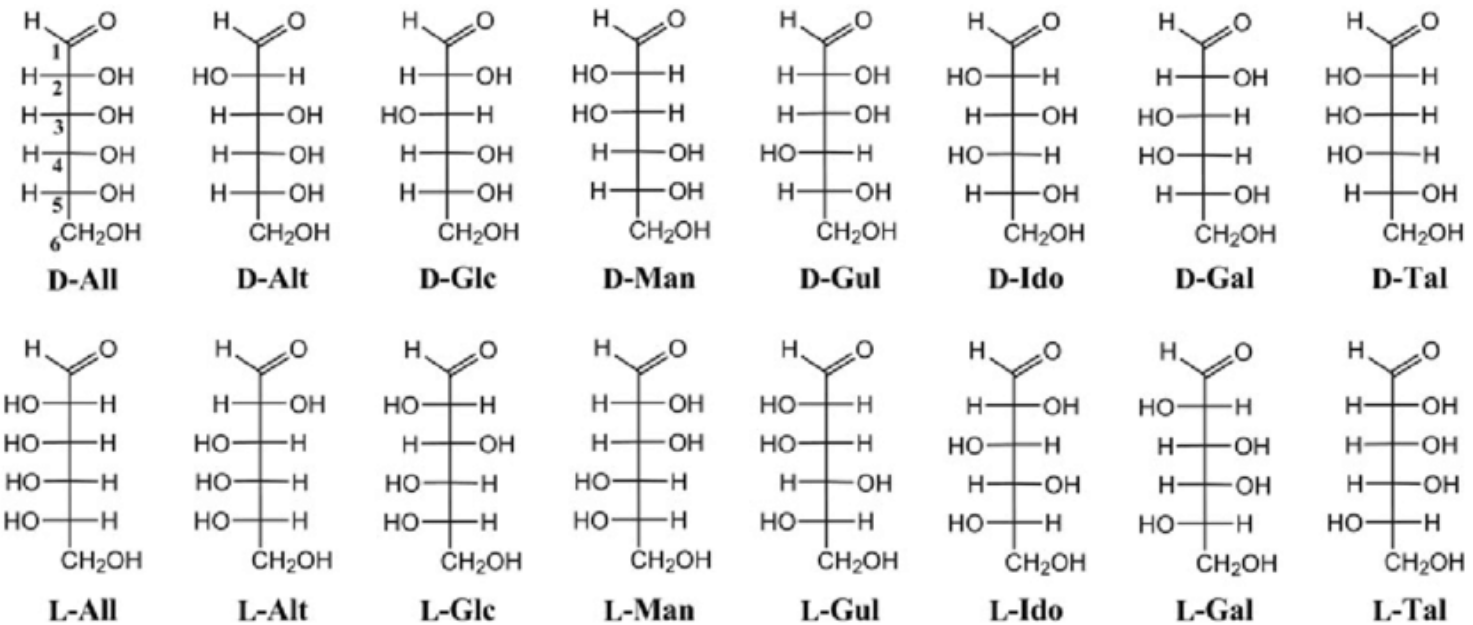
Part 2

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How many different stereoisomers are possible for linear aldohexose ?



- $n = \text{four chiral (*) carbons}$
- $\text{Number of stereoisomers} = 2^n = 2^4 = 16$



How many disaccharides are theoretically possible with two Glucopyranose ? How many of them Are Non-reducing sugars ? (NB: Anomeric carbon must be involved in GLYCOSIDIC linkage).

➤ Stereochemistry (consider D if not mentioned)

➤ Pyranose = six membered ring

α -Glucose + α -Glucose

α -(1→1)- α

α -(1→2)

α -(1→3)

α -(1→4)

α -(1→6)

β -Glucose + α -Glucose

β -(1→1)- α → SAME as α -Glucose + β -Glucose

β -(1→2)

β -(1→3)

β -(1→4)

β -(1→6)

α -Glucose + β -Glucose

α -(1→1)- β

α -(1→2)

α -(1→3)

α -(1→4)

α -(1→6)

β -Glucose + β -Glucose

β -(1→1)- β

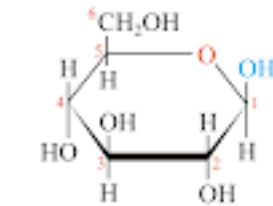
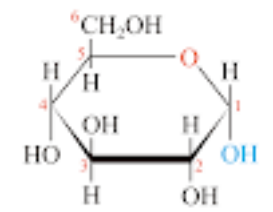
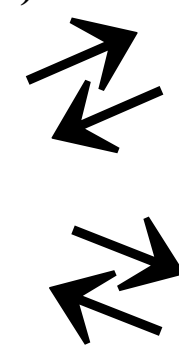
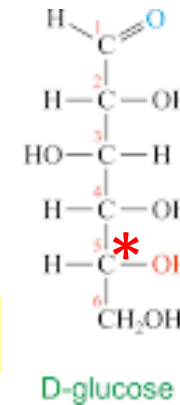
β -(1→2)

β -(1→3)

β -(1→4)

β -(1→6)

TOTAL= 20 → 19



D-Glucopyranose

Non-reducing disaccharides = (α - α , α - β , β - α , β - β) = 4 → 3

25 mg sample of *E. Coli* glycogen was completely hydrolyzed in H_2SO_4 .

The neutralized final mixture is 10 ml. The glucose content of the final solution was 2.35 mg/ml.

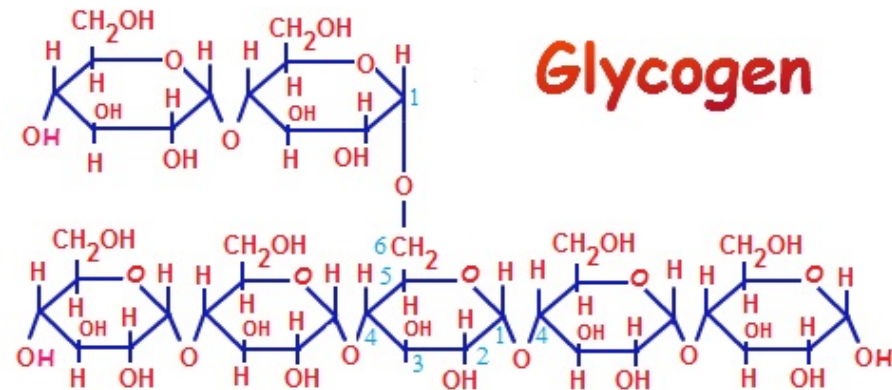
What is the purity (in %) of the isolated glycogen ?

- Total amount of glucose obtained after hydrolysis = $2.35 \text{ mg/ml} \times 10 \text{ ml} = 23.5 \text{ mg}$
- Molecular wt of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) = $(6 \times 12 + 12 \times 1 + 6 \times 16) = 180 \text{ mg/mmol}$
- Total amount of glucose after hydrolysis = $(23.5 \text{ mg} / 180 \text{ mg.mmol}^{-1}) = 0.1306 \text{ mmol}$

Glycogen sample contain 0.1306 mmol of glucose.

But when glucose is polymerized an H_2O is removed to form glycosidic bond.

THUS, Glycogen = $(\text{C}_6\text{H}_{10}\text{O}_5)_n$

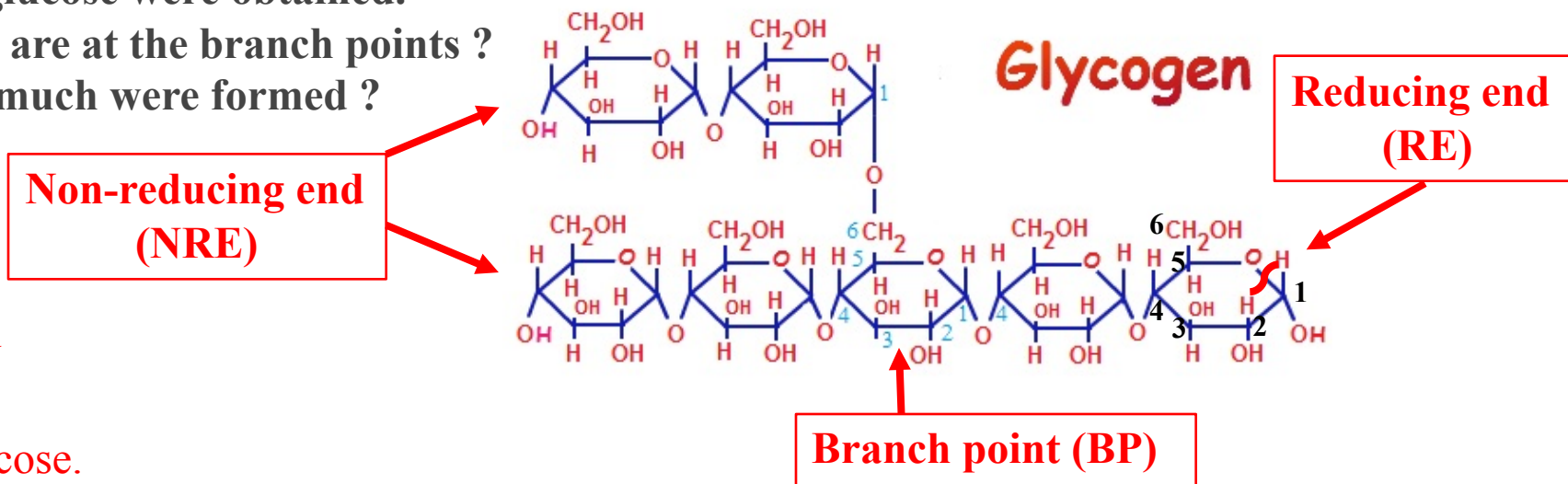


- Thus, MW of glucose in glycogen = $(\text{C}_6\text{H}_{10}\text{O}_5) = (6 \times 12 + 10 \times 1 + 5 \times 16) = 162 \text{ mg/mmol}$
- Thus, 0.1306 mmol glucose corresponds to = $(0.136 \text{ mmol} \times 162 \text{ mg/mmol}) = 21.16 \text{ mg GLYCOGEN}$

$$\text{PURITY} = (21.16 / 25) \times 100 = 84.6 \%$$

81 mg of human glycogen were exhaustively methylated and then acid hydrolyzed. The products were separated and identified. 62.5 μmoles of 2,3-dimethyl glucose were obtained.

- (a) What percentage of glucose residues are at the branch points ?
- (b) What were other products and how much were formed ?



- Each free OH- group can be methylated
- Branch-point (BR) ➔ 2,3-dimethyl glucose.
- Total glucose in 81 mg glycogen = $(81 \times 10^{-3} \text{ g}) / (162 \text{ g/moles}) = 5 \times 10^{-4} \text{ moles} = 500 \text{ μmoles}$
- **2,3-dimethyl glucose** comes from the branch points => **% BP = $(62.5 / 500) \times 100 = 12.5 \%$**
- If, BP = n residues, then NRE = n+1 residues. But if n is large, then n+1 ~ n.
- Methylation of NRE will produce 2,3,4,6-tetramethyl glucose.
- **2,3,4,6-tetramethyl glucose => 62.5 μmoles**

Methylation of RE will produce 1,2,3,6-tetramethyl glucose. But upon hydrolysis the methyl group of Carbon-1 will be released as methanol. i.e, 1,2,3,6-tetramethyl glucose ➔ 2,3,6-trimethyl glucose.

- **THUS TOTAL 2,3,6-trimethyl glucose = $(500 - 2 \times 62.5) = 375 \text{ μmoles}$**

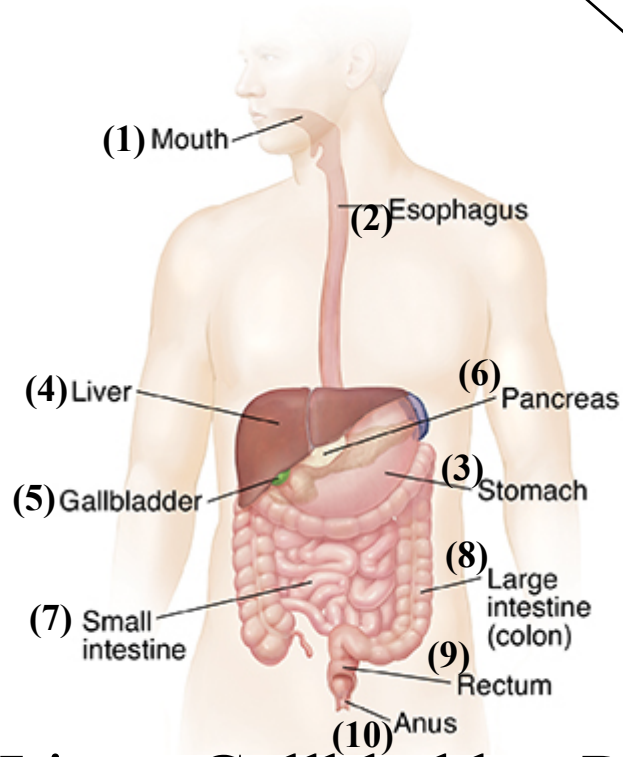
Human food intake $\sim 1 - 2.7$ kg / day
 ~ 365 kg/ year
 ~ 29200 kg/ lifetime (80 years)

Food makes its way through **Digestive System**

4 main components

Digestive System

[1]. Gastrointestinal track (See Figure below 1,2,3,7,8)



Internal surface area 30-40 m²
> 1/2 of a badminton court



[2]. Liver, Gallbladder, Pancreas (see Figure above 4,5,6) :
Special juices produced and mixed with food

[3]. Enzyme +Hormones+Nerves+blood for breaking down food, product delivery

[4] **Mesentery**: Large stretch of tissue that support and positioned all the digestive organs in the abdomen.

Body = All about balance



Homeostasis : A state of balance in the body

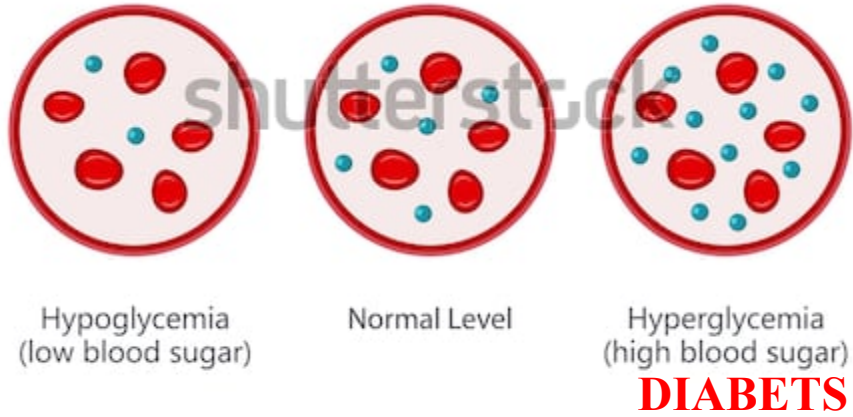


Body Temperature = 37°C

Blood Glucose = 80 – 100 mg/dL (fasting)

Imbalance = Disease

BLOOD GLUCOSE LEVELS



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BLOOD SUGAR LEVEL CHART

	FASTING	JUST ATE	3 HOURS AFTER EATING
NORMAL	80-100	170-200	120-140
PRE-DIABETIC	101-125	190-230	140-160
DIABETIC	126+	220-300	200+

Blood glucose shoot up after your meal

Blood glucose shoot up differently for different food...

Ranking carbohydrates → Glycemic index (GI)

High GI food

→ Rapidly digested and absorbed

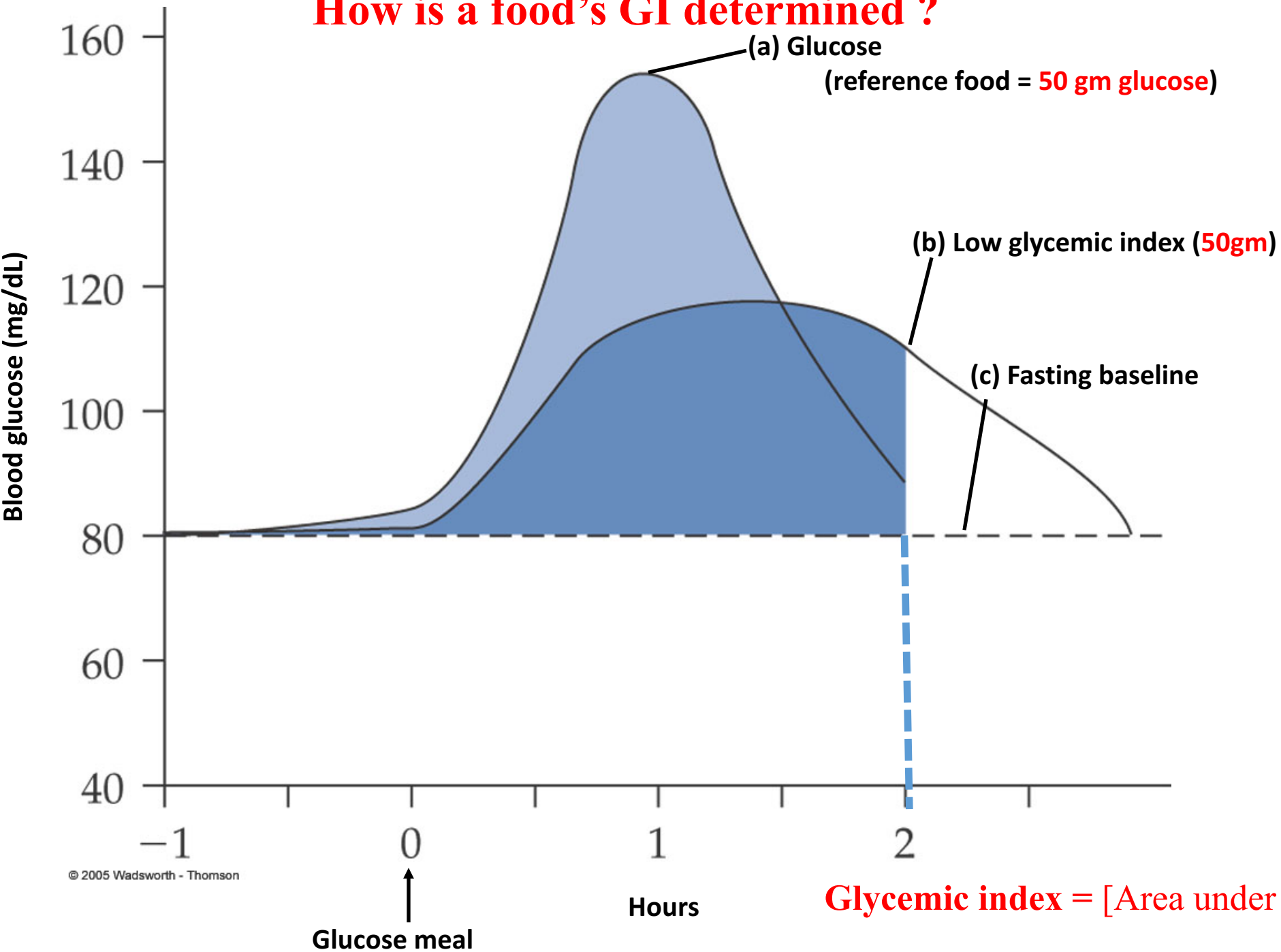
→ Shoot up blood glucose level.

Low GI food

→ slowly digested and absorbed

→ gradual increase blood glucose level.

How is a food's GI determined?



Glycemic Index Classification

GI

GI Rating

Effect

HIGH
GI

≥ 70

Rapid increase in
blood sugar levels

MEDIUM
GI

56-69

Moderate increase in
blood sugar levels

LOW
GI

≤ 55

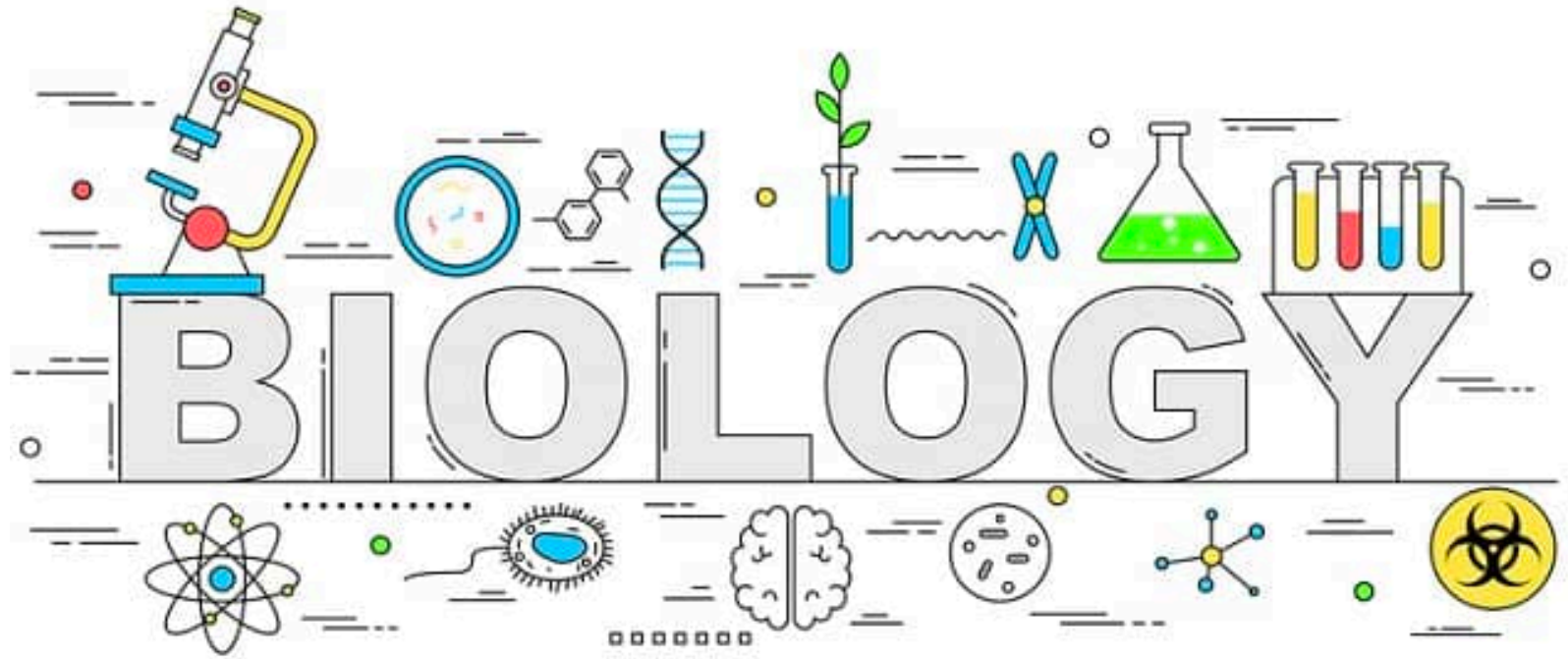
Slow increase in
blood sugar levels

Glycemic index

- Useful for blood sugar level spikes
- Low GI does not mean Healthy/Unhealthy

Should be considered

- Fiber content
- Fat content
- Cooking procedure
- Food processing
- Etc etc...



Next: *Sugar-Dark Side of Sweetness*